

Ocean Guard AI

An AI-powered system for marine protection and research



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I Project Description

1 Project Overview

Ocean Guard AI is a large-scale, AI powered system designed to monitor and protect marine life in real time. It utilizes underwater drones, sonar arrays, and advanced machine learning models to detect and classify threats to marine life across the ocean. When a threat is identified, the system can alert the user enabling them to trigger safe deterrents (sound pulses or light signals) and collect valuable data for research. In addition to immediate response, Ocean Guard AI will guide a global archive of marine life interactions. This will support long-term studies on species behavior, migration as well as the impacts of climate change.

2 The Purpose of the Project

The purpose of Ocean Guard AI is to provide scientists, conservationists, and policy makers with a reliable, real-time tool for protecting vulnerable marine species and ecosystems. By combining live detection with advanced analytics, the system will not only give endangered species a chance at survival during dangerous encounters but will also generate insights that can guide conservation strategies and inform global research efforts.

2a The User Business or Background of the Project Effort

The main users of our technology are marine biologists. The “business” of this project is global marine conservation. Today, marine ecosystems face threats from pollution, climate change, overfishing, and human marine activity. Organizations such as marine research institutes, government agencies, NGOs, and fisheries are actively seeking new ways to monitor and protect ocean life. However, there is currently no real-time global monitoring system capable of detecting predator-prey interactions or providing automated responses at scale.

With Ocean Guard AI, users will be able to:

- Receive real-time alerts when vulnerable species are under threat.
- Activate safe deterrents to reduce harm without disrupting ecosystems.
- Collect long-term, reliable datasets for studies on migration, population trends, and climate change.
- Integrate findings with satellite imagery and GPS tracking for a more complete understanding of marine ecosystems.

Without a system like Ocean Guard AI, vulnerable species remain at heightened risk due to human impacts and environmental change. Developing this platform provides both immediate protective value and long-term research benefits, addressing urgent conservation needs while also laying the foundation for global scientific collaboration.

The problems faced by marine conservation are serious and urgent. Species continue to decline because current methods are too slow and limited. Without a global, real-time system, conservation teams cannot respond fast enough to threats or gather the

data they need for long-term solutions. This shows the problem is significant and must be solved.

2b Goals of the Project

The goal of Ocean Guard AI is to provide a real-time, reliable, and scalable monitoring and protection system for marine ecosystems. The project aims to improve the survival chances of vulnerable marine species by giving marine biologists, conservationists, and governments actionable tools that directly enhance their work. The ultimate purpose is not just to create software, but to enable effective conservation, real-time protection, and valuable long-term ecological research.

As marine species face growing threats from human activity and climate change, current monitoring systems are too slow, localized, or reactive to make a timely difference. By setting clear goals, Ocean Guard AI avoids drifting into purely technical development and stays focused on the higher mission: protecting biodiversity and strengthening global conservation.

We want to provide marine scientists with immediate alerts when a species is under threat. We also want to allow conservation teams to intervene in real time using safe deterrents. For example, redirecting a turtle away from a polluted area using a sound deterrent.

2c Measurement

To know if our goals are met, we need clear ways to measure them. For example, one goal is that the system should detect predator-prey encounters with around 90% accuracy. We can test this by using sample data that has already been labeled by experts and then comparing our results.

Another goal is that the system should respond quickly, within 5–10 seconds, and we can measure this by checking how long it takes from detection to when an alert or deterrent is actually triggered.

To see if the system really helps animals, we can compare survival rates in situations before and after using Ocean Guard AI. For research, we will look at how many useful datasets are created and how many studies or reports end up using them.

Finally, to know if the system is being adopted, we can count the number of organizations like NGOs, governments, and fisheries that are using Ocean Guard AI within the first three years. It is important that these measurements are set early, because they give us a clear way to prove success and keep the project focused on its original goals.

3 The Scope of the Work

The business of this project is marine conservation, but the work of Ocean Guard AI is focused specifically on monitoring predator-prey encounters in real time and helping conservation teams respond quickly. Its role is limited to detection, alerts, deterrents, and data collection, not the entire scope of ocean conservation.

3a The Current Situation

Right now, marine conservation work is mostly done through manual observation, tracking tags on animals, and occasional surveys. These methods provide useful data,

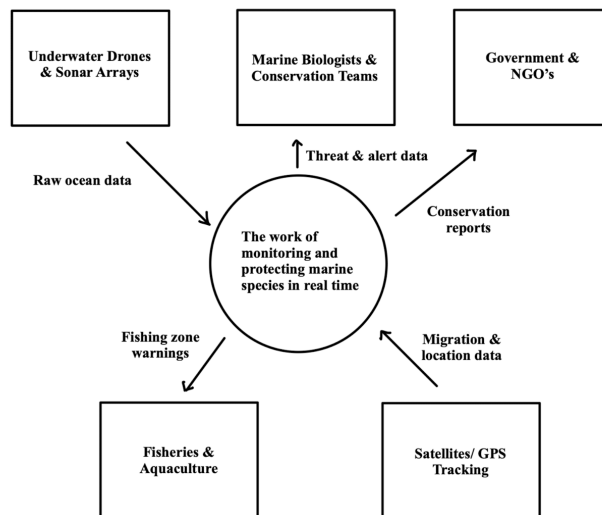
but they are slow, limited to small areas, and often analyzed long after the events happen. There is no global system that can monitor threats in real time.

Because the current methods are delayed and incomplete, conservation teams often cannot act quickly enough to prevent harm to vulnerable species. By understanding these limitations, we can see why a system like Ocean Guard AI is needed to fill the gap and give faster, more reliable information.

3b The Context of the Work

The work of Ocean Guard AI is limited to monitoring predator-prey encounters, sending alerts, activating safe deterrents, and storing data for research. It does not handle larger areas of marine conservation, like setting policy or enforcing laws. To work properly, the system will need to interact with outside groups and data sources, such as marine biologists who receive alerts, government or NGO agencies that use the data, fisheries that adjust their practices, and satellite/GPS systems that provide extra environmental information.

By defining these boundaries, we make it clear what Ocean Guard AI is responsible for and what it depends on from outside systems. This helps keep the project focused and ensures the product will fit cleanly into the larger marine conservation environment.



3c Work Partitioning

Below is the list of the main business events Ocean Guard AI responds to. Each event shows the input from adjacent systems, the output, and a short summary of the business use case.

Event Name	Input	Output	Summary
Drone sends ocean data	Raw ocean data from drones/sonar	Stored data for analysis	System ingests and processes ocean data

Predator-prey encounter detected	AI classification of species behavior	Threat alerts to biologists	System detects a threat and alerts conservation teams
Deterrent activated	Operator approval or automated rule	Safe deterrent signal	System triggers a deterrent to protect the species
Satellite sends migration data	Satellite/ GPS tracking data	Updated migration maps	System refined risk zones with satellite info
Weekly report due	Scheduled task	Conservation reports to NGOs	System generates summary reports for stakeholders

These events represent the major real-world triggers that Ocean Guard AI must respond to. By breaking the work into these smaller parts, we can clearly define the system's responsibilities and later turn them into specific requirements.

3d Competing Products

There are some technologies that do parts of what Ocean Guard AI would do, but none provide the same full solution. For example, there are existing marine monitoring systems that use sonar arrays or GPS tagging to track specific species, but these are usually limited to small study areas and do not operate in real time. Drones are also used in marine research, but mainly for short missions or local observation. Some AI models are being tested for identifying species in images or videos, but they are not connected into a large network for global monitoring. None of these existing products combine drones, sonar, and AI with safe deterrents in a single, integrated system.

By looking at what already exists, we can see that Ocean Guard AI fills an important gap. Current tools are good for collecting data, but they cannot stop threats in the moment or provide a worldwide monitoring and protection system. Our project is justified because it adds real-time detection, immediate response, and large-scale integration that other products do not offer.

Existing products may still be useful as components or inspiration (for example, using current tagging systems or existing AI models as part of training). However, they are not complete solutions for marine protection, which is why Ocean Guard AI is needed.

4 The Scope of the Product

The scope of Ocean Guard AI is to serve as an intelligent monitoring and response system for marine conservation. It does not attempt to regulate entire ecosystems or replace conservation policy, but instead focuses on four key functions: detection, alerts, safe deterrence, and analytics.

For example, a marine biologist studying manatees could receive an instant alert when a shark approaches a herd, saving hours of manual video review. A conservation team monitoring sea turtles during migration could see predator hotspots and activate

deterrents at the right time and place. International research institutions could share data through the system’s global database, publishing joint studies that would not be possible with isolated data. Fisheries organizations could use predator activity reports to adjust fishing zones responsibly, protecting both biodiversity and their own industry’s sustainability.

These scenarios illustrate the range of users and benefit that Ocean Guard AI will provide. Its scope is focused but broad enough to support conservation, research, and responsible resource management.

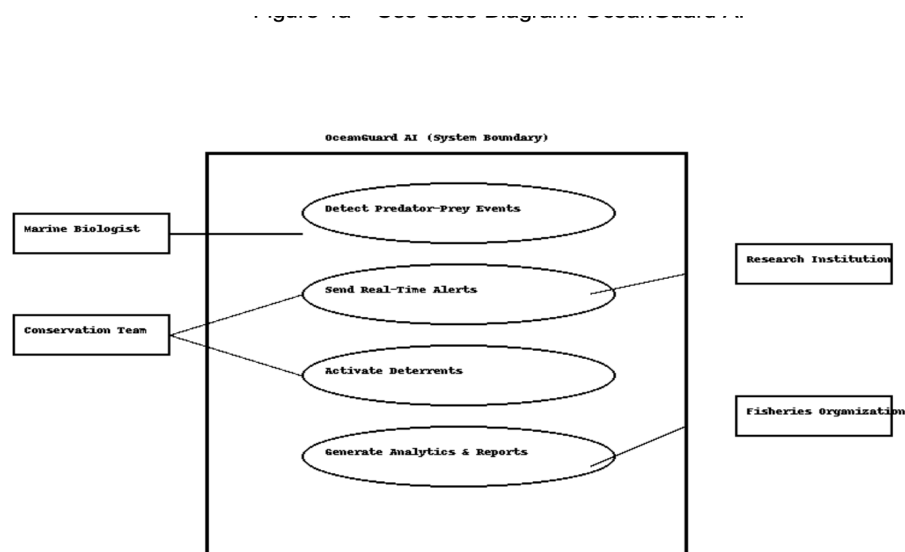
4a Scenario Diagram(s)

The scenario diagram illustrates the relationship between Ocean Guard AI and its external actors. On the outside of the system boundary are marine biologists, conservation teams, fisheries organizations, research institutions, and the marine environment itself, which provides the raw data. Inside the boundary, Ocean Guard AI processes detection, alerts, deterrents, and analytics. Arrows connect each external actor to the features they rely on, showing how the system supports real-world use. The use case diagram identifies the system’s primary functions: detecting predator-prey events, sending real-time alerts, deploying safe deterrents, and providing analytics and reports. Each external actor connects only to the functions they depend on, clarifying the boundaries of the product.

4b Product Scenario List

Ocean Guard AI supports several distinct scenarios:

Marine Biologist in the Field, Marine biologist, sensor/drone network
 Conservation Team Monitoring Species, Conservation team, sensor/drone network, deterrent devices
 Global Research Collaboration, Research institutions, marine biologists
 Fisheries Management Application, Fisheries organizations, research institutions.



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Marine Biologist in the Field, Marine biologist, sensor/drone network Conservation Team Monitoring Species, Conservation team, sensor/drone network, deterrent devices Global Research Collaboration, Research institutions, marine biologists Fisheries Management Application, Fisheries organizations, research institutions.

Conservation Team Monitoring Species: A turtle conservation team uses sonar buoys linked to Ocean Guard AI. At night, the system detects predators near the turtles and suggests activating safe deterrents. The team reduces losses and later reviews a heatmap of high-risk areas.

Global Research Collaboration: Researchers in Canada and Japan upload data to the system's shared database. The AI highlights similar predator strategies across oceans, leading to a joint publication and advancing global science.

Fisheries Management Application: A fisheries group in Chile uses Ocean Guard AI's weekly reports to track predator activity. When predators cluster around anchovy schools, they adjust fishing zones to protect stocks and preserve sustainability.

5 Stakeholders

The Client

The client for Ocean Guard AI would likely be a marine conservation organization or government agency such as WWF, NOAA, or UNEP. These groups provide funding and strategic direction, motivated by protecting biodiversity and advancing conservation technology.

The Customer

Customers include conservation groups, government agencies, fisheries organizations, and research institutions. They would purchase Ocean Guard AI to gain more effective tools for real-time monitoring, long-term reporting, and resource management.

Hands-On Users

Hands-on users include marine biologists, conservation teams, fisheries managers, and research scientists. Each group has different expertise and technology skills but shares the need for accurate detection, usable dashboards, and actionable data. Marine biologists and researchers contribute scientific expertise, while conservation teams rely on usability in the field. Fisheries managers focus on sustainability through analytics.

Maintenance Users

Maintenance users include system administrators, developers, and field technicians who install, update, and service the system. They require robust documentation, modular design, and durable hardware.

Other Stakeholders

Other stakeholders include sponsors, testers, business analysts, technology experts, legal advisors, and usability experts. International associations such as UNEP or IUCN also have a role in ensuring that the system fits into global conservation efforts.

Recognizing these groups ensures that requirements are not missed and that the system gains broad support.

User Participation

Users must be actively involved throughout development. Marine biologists and conservation teams will test AI accuracy and usability in the field, while fisheries and research institutions will shape reporting standards. Maintenance users will define requirements for sustainability. Without this participation, requirements risk being incomplete or misaligned.

User Priorities

The most important users are marine biologists and conservation teams, as they depend on the system for daily conservation work. Research institutions and fisheries organizations are secondary users whose input is valuable but not critical to core success. The public may benefit indirectly, but their needs are not prioritized in design.

6 Mandated Constraints

However, building this Ocean Guard AI system, we will face certain obstacles that can halt or slow down the progression of the development. We have to account for multiple different constraints including technology functionality, environmental challenges, device connections, budgeting of the project, and the limited time of development. We have taken the constraints that will come with developing the project into account and thought of some ideas to resolve these conflicts.

6a Solution Constraints

First, we need to think about the data collecting devices that we plan on using for this project. We have underwater drones to collect data about marine life which also plays a role in correctly identifying and classifying certain species in the ocean. They will observe and gather information about marine life, like their homes, diets, predators, and lifestyle.

Now these drones will contain AI models and algorithms to manage the data collection and management in the databases of the system. The constraints about this are the accuracy of these AI models. It needs to be made sure that these AI models are collecting and identifying information correctly regarding threats to their environment and way of life. Misidentifying certain threats under the system could lead to false positives and ineffective safe deterrents due to misinterpretation of the AI models. These AI models need to be investigated consistently for accurate information gathering.

The networks of the oceans are unstable. This means that our underwater drones may need to operate offline, instead of online Wi-Fi-dependencies. This is mainly because there are more challenges of communication and limitation of wireless signals underwater. We need to establish other forms of network functionality offline to prevent loss of drone control, delayed responses, malfunctioning GPS navigations, and low radio connection. Operating under Wi-Fi networks, our drones are highly at risk of losing connection to our control system, mapping navigations, and even losing potential data acquired by those specific drones.

Importantly to note as well, the storage capacity of these drones needs to be reasonable for these huge amounts of data collection. We don't want to miss out on our data collections because the drones ran out of storage space. Under the ocean depths, the transferring of data to the system might require higher speeds of communication and connection. The computing of these underwater drones needs to be able to perform and function properly under the unstable network connection in these environments. We could have trouble retrieving real-time data that is up to date and cause delays in sending real threat alerts when these are not accounted for.

6b Implementation Environment of the Current System

The marine conditions could create problems for our underwater drones. The further down these underwater drones go into the ocean abyss, the more pressure they would withhold. With a large amount of underwater pressure, the hardware of the underwater drones has to be shielded from pressure to prevent hardware damage to our data collectors. They must be built with pressure-resistant and waterproof capabilities. Internal components under harsh pressures of the ocean can implode due to high amounts of pressure.

The temperature of the ocean depth holds potential threats to the drone's battery life. Though these drones are designed for underwater environments, extreme water temperatures can still affect the electronics, battery, and sensors of these machines. This is kind of like how aerial drones are affected by air temperature regarding battery efficiency and electronic functionality. We would require our underwater drones to have a reasonable battery life that will allow them to function underwater for long periods of time. We want these drones to operate more than 24 hours underwater as extreme cold temperatures can drain battery life.

Salinity and corrosive factors have potential effects on underwater drones that need to be resolved. The salinity (Salt Concentration) and other corrosive factors boost up the degradation of materials embedded into our drones. Degradation of materials can cause electrical failures, hardware issues, and low structural capacity against high pressure. High amounts of salinity or salt in the water increase the amount of electrical conductivity of the device. With high amounts of electrical conductivity, the water conducts more electrochemical corrosion turning the water environment into one aggressive electrolyte that facilitates even more corrosive factors.

6c Partner or Collaborative Applications

The collaboration and partnering of NGOs and research institutions are vital for this project's progression. Partnering with research institutions and NGOs, we can acquire expert validated data regarding underwater wildlife and expand coverage globally. It can provide technological development and resources to help develop the needs of the system. Because we see our stakeholders to include expert researchers on marine biology and conservation, Ocean Guard AI is something that these research institutions can use.

When implementing Ocean Guard AI, we need to be mindful of global policy on environmental principles. It should be assured that Ocean Guard AI is compliant to underlying regulations of collective actions, rules, and agreements to address environmental challenges. Governmental agencies like the US Environmental Protection Agency and the National Oceanic Atmospheric Administration play a key role in enforcing laws and regulations when it comes to developing things that might

interfere. Ocean Guard AI needs to account for any form of unethical use of the environment, especially if it violates environmental regulations.

It would also be helpful if we could get Ocean Guard AI access to satellite networks. Because underwater drones and sonar arrays only cover a tiny, limited area, the use of satellites can help expand the system's coverage area. Satellites allow GPS tracking of marine animals and migration patterns in distances beyond our devices' reach. Satellites work as relay channels, where they acquire data from one destination then transport it to another destination over a vast distance. It allows our underwater drones to transmit collected data faster to labs and the system.

In relation to satellites, they collect information on marine life analysis including sea surface temperatures, food availability, and pollution levels in ecosystems. As many conservationists have already used satellites to track marine animals, integration of Ocean Guard AI is a good step to expanding it.

6d Off-the-Shelf Software

Due to the large-scale data storage, the monitoring system and drones will rely on, we plan to use PostgreSQL, Docker, and TensorFlow. However, constraints to this might come with compatibility and scalability issues. Not all software, including the ones listed, work smoothly with underwater drone devices and sonar arrays especially if it is done in a single data management system/. They might require more than just one software to function properly and assume additional custom adapters and APIs. PostgreSQL and Docker have limitations to their scalability where handling more massive data in a large area might go beyond what they can handle when collecting real time data.

As mentioned before, the network connections in underwater environments are unstable. This creates problems with our planned software as they might assume a kind of internet Wi-Fi or cloud access function. As we have talked in the previous paragraphs that Ocean Guard AI might need to operate in an offline setting, we need to find a way that the off-the-shelf software we are using can operate in the respective setting. This is something that software platforms may not fully support and might be challenging to make it work without the internet.

Importantly, sensitive data especially from research and governmental institutions needs to be safeguarded with the planned software that we are using. The Ocean Guard AI system needs to be protected from cyber-attacks and hackers. Because data in satellite and underwater drones is transmitted wirelessly, they can be intercepted and compromised. Privacy is another thing that needs to be accounted for, especially policy regulations in data protection.

6e Anticipated Workplace Environment

For Ocean Guard AI, it will work in multiple work environment collaborations. The workplace environments under Ocean Guard AI are a mix of labs, ships, research facilities, and offices for human workers. Then we also have our device-controlled workers like drones, sonar arrays, and satellites combining the efforts of underwater, surface level, and space. Now this would make the scale of the project very large. The splitting of the work environments into many can create more time in development

because software and hardware's used would have to be tested in both controlled conditions of the labs and uncontrolled conditions at sea.

In return we would also have to gather a large number of professionals in different professions to develop the perfect marine life monitoring system. This would take the combined work of multiple different teams all collaborating on one goal. In our case, the following teams being considered would be Research, Programming, Management, Data Analytics, Engineering, Network, Environmental Policy, Interface Designers, and Ship Crews.

The research team consists of marine biologists, ecologists, and conservationists actively researching and studying marine life. The programming team will deal with developing the backend of the systems as well as the AI models used to classify marine life. The management team is responsible for resource management and budgeting of the project. Data analytics will consist of data scientists and analysts analyzing and tracking the datasets being collected. The engineering team will be a group of robotics and hardware engineers building the devices and keeping them properly functioning. The network team will oversee establishing network solutions so that satellites, sonar arrays, and underwater drones can communicate with one another effectively even in unstable situations. The environmental policy team will be responsible for keeping the system in check in terms of regulations and compliance with laws. Interface designers will deal with building the interfaces for Ocean Guard AI to build dashboards and easy to interpret visualizations. Lastly, we have our crew members operating in ships often deploying underwater drones and using sonar arrays to gather information.

The sheer number of different professions Ocean Guard AI needs would require more emphasis on interdisciplinary teamwork of professions, instead of a single software project. Safety protocols would also need to be initiated in cases of bad weather situations in the sea for ship crew workers. Safety regulations can also limit our testing grounds for deploying sonar arrays and drones, which is why the environmental policy team is important. Because our workers would be so spread out, we would also need to account for different time zones and the effects that they can have in decision making.

6f Schedule Constraints

As you can see, the development of Ocean Guard AI is a long-term development project adding some time constraint to it with expected deadlines. The process of systems programming, technological readiness, environmental testing, and data collecting would consume a lot in a span of limited time to build the most we can of Ocean Guard AI. Because these systems cannot be fully built and ready to go all at once, it is best to break up the development into multiple different stage phases. These stage phases must be completed in chronological order, but with deadlines to get the most we can for the whole semester.

To add as well, because the separate locations of multiple teams working collaboratively in labs, ships, offices, and research facilities would be so spread out, the scheduling might take a bit more effort considering that the time zones will not be the same for every worker for Ocean Guard AI. The cross-team collaboration of developers, marine biologists, and even the partnering institutions would have to synchronize scheduling plans and testing periods. In consequence, this could probably slow down communication and approval by a lot.

Importantly, the testing periods and deployment of these data collecting devices and sonar arrays would have to be really well planned out. First these devices needed to be tested before being actually used and faced potential delays in their development due to environmental challenges, functionality issues, inaccuracy, a limited access to testing vessels, and environmental regulations. After testing, the scaling to additional regions, refining AI accuracy, and securing data infrastructure would be carefully thought out before actually implementing them.

6g Budget Constraints

With Ocean Guard AI, a large-scale project, would require an immense amount of budget due to developing underwater drones, AI models, satellites usage, global data structures, and multiple teams of different professions. Because of the size of the work environment and resources, we estimate that a fully functioning Ocean Guard AI system would cost millions of dollars to create, around \$5 million to \$15 million to create. This is one of the major critical constraints to our project.

The manufacturing of the underwater drones and sonar arrays would be really expensive. Implementing hardware that is resistant to pressure and implosion plus the harsh temperatures of the depths, we would need high-quality materials and precise engineering. The development of the AI models would require a lot of money to train and maintain so that they can process data in unstable underwater connections, fast computation, and large data storages.

Then there comes the cost of developing our labs, research facilities, and ships. The need for field testing, renting ships, acquiring the necessary equipment and logistics of deploying drones would add a couple hundred thousand more to the budget. Because we are considering partnering with research institutions, organizations, and services, we would need to account for the fees of collaboration and accessing their resources.

7 Naming Conventions and Definitions

7a Definitions of Key Terms

Entity	Attribute	Definition
Species	SpeciesID	Unique identifier for each species
	Name	Common name of the species
	ScientificName	Scientific classification name
	Predators	Species that prey on this species
	Prey	Species consumed by this species

	Diet	Dietary type (carnivore, herbivore, omnivore)
	RiskLevel	Endangerment or protection status
Encounters	EncounterID	Unique identifier for encountering events
	SpeciesInvolved	List of species participating in the encounter
	LocationOfEncounter	Geographic location where the encounter occurred
	TimeOfEncounter	Timestamp of the encounter
	TypeOfEncounter	Timestamp of the encounter
	Description	Additional details or notes
Alerts	AlertID	Unique identifier for each alert
	EncounterID	Reference to related encounter
	PriorityLevel	Urgency level of the alert
	SpeciesThreatened	Species at risk in the alert
	SpeciesThreat	Species or factor causing the threat
Drones	DeviceID	Species or factor causing the threat
	DeviceType	Type of drone
	DeviceLocation	Geographic coordinates or assigned location
	DataCollected	Information gathered (video, sonar, environmental)
	DeviceStatus	Operational status
Monitors	DeviceID	Unique identifier for each sonar/ sensor device
	NetworkCoverageArea	Geographic area monitored by the device

	DeviceLocation	Placement of the device
	DeviceSensitivity	Sensitivity level of detector
	DeviceStatus	Operational status
	DetectedEncounters	List of encounters detected
	DetectedMovements	Movement patterns detected
	Description	Additional notes
Users	UserID	Unique identifier for each user
	UserProfession	Role or profession of the user
	PhoneNumber	Contact phone number
	Email	Contact email address
	StartDate	Date when the user registered in the system

7b UML and Other Notation Used in This Document

The system uses UML notation to describe components, actors, and data flows. Arrows represent data transfer, actors represent users (e.g., biologists, NGOs), and the system boundary outlines Ocean Guard AI's scope. Consistent notation ensures all diagrams are interpreted the same way by different stakeholders. If notation is only used once (e.g., in a specific diagram), it should be explained directly in that section. This document follows the OMG UML 2.0 standard, with exceptions noted where applicable.

7c Data Dictionary for Any Included Models

The data dictionary explains the main information Ocean Guard AI uses. For species, we keep track of IDs, names, diets, and risk levels. Encounters store details like which species were involved, where and when it happened, and what type of interaction occurred. Alerts connect to encounters and show the urgency, the species at risk, and the cause of the threat. Drones and monitors are described by their IDs, location, status, and the data they collect. Finally, users are recorded with their ID, role, and contact information.

The dictionary makes the system's information clear, so everyone knows exactly what each piece of data means. This avoids confusion and keeps the project consistent. The data dictionary helps both analysts and developers. Analysts define the important terms, while developers can later add the technical details when building the system.

8 Relevant Facts and Assumptions

8a Facts

Important background information for the Ocean Guard AI project:

- No real-time global monitoring system exists today. The world lacks operational systems which enable real-time observation of marine life.
- Marine biologists, conservationists, and governments are actively looking for technologies that protect marine life.
- The world needs technological solutions which will help protect marine ecosystems and preserve biodiversity.
- Climate change and human impacts have increased biodiversity loss over the years. The rapid decrease in marine biodiversity requires immediate action for Ocean Guard AI development because of its critical importance.

8b Assumptions

The following assumptions are made to guide the system design:

- Under harsh ocean conditions, underwater drones and sonar arrays will operate properly and effectively. It is assumed that the system's hardware can withstand high pressure, salinity, and other challenging marine environments.
- AI models demonstrate the ability to extract precise and dependable information for species classification and identification tasks. The system will generate precise information which researchers and conservation teams can use for their work.
- Global regulations and ethical compliance regarding environmental protection and endangered species will be satisfied. The project follows all international laws and regulations and ethical standards that protect marine life.

